Arista Access Node WFD-ACC-14 Major Failure SOP for Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss)

Node: WFD-ACC-14 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_4747

Description

This SOP addresses a major failure scenario on the Arista Access Node WFD-ACC-14 at the Whitefield POP, involving Hardware/Environmental Faults (ASIC memory parity errors, line card crashes, power supply redundancy loss). These failures are attributed to the inherent characteristics of the Arista 7150 series switches, which utilize custom-designed ASICs. The line cards, which are crucial for forwarding traffic, can crash due to environmental factors such as overheating, power supply redundancy loss, or ASIC memory parity errors. The Arista 7150 series switches also employ a distributed architecture, where each line card operates independently, which can lead to isolated failures. The ASIC memory parity errors can occur due to manufacturing defects or environmental factors like temperature fluctuations. The power supply redundancy loss can be caused by a faulty power supply unit or a failure in the power distribution system. The Arista switches also employ a redundant power supply system, which can fail to provide adequate power to the line cards, leading to crashes. This SOP outlines the diagnostic, resolution, and rollback procedures for addressing these failure scenarios.

Pre-requisite Checks

- Verify the redundant link status on the Arista Access Node WFD-ACC-14.
- Ensure that the remote hands are dispatched to Whitefield POP to perform physical checks.
- Check the environmental conditions at the site, including temperature and humidity levels.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to manufacturing defects or environmental factors like temperature fluctuations.	High
Line card crashes due to power supply redundancy loss or overheating.	Medium
Power supply redundancy loss due to faulty power supply unit or failure in the power distribution system.	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal power, Expected Output: Power supply status: OK, Description: This command checks the power supply status of the Arista Access Node WFD-ACC-14. The output indicates whether the power supply is functioning correctly or not.
```

```
Command: show system internal temperature, Expected Output:
Temperature: 40°C, Description: This command checks the temperature of
the Arista Access Node WFD-ACC-14. The output indicates whether the
temperature is within the recommended range or not.

Command: show system internal asic memory, Expected Output: ASIC
memory status: OK, Description: This command checks the ASIC memory
status of the Arista Access Node WFD-ACC-14. The output indicates
whether the ASIC memory is functioning correctly or not.
```

Resolution Steps

Step 1: Step: Replace the faulty power supply unit., Command: replace power supply unit 1, Description: This step involves replacing the faulty power supply unit with a new one. The command `replace power supply unit 1` replaces the first power supply unit with a new one.

Step 2: Step: Replace the faulty line card., Command: replace line card 1, Description: This step involves replacing the faulty line card with a new one. The command `replace line card 1` replaces the first line card with a new one.

Step 3: Step: Reset the Arista Access Node WFD-ACC-14., Command: reset system, Description: This step involves resetting the Arista Access Node WFD-ACC-14 to its default configuration. The command `reset system` resets the system to its default configuration.

Post-Check Verification

```
Command: show system internal power, Expected Output: Power supply
status: OK, Description: This command checks the power supply status
of the Arista Access Node WFD-ACC-14. The output indicates whether the
power supply is functioning correctly or not.

Command: show system internal temperature, Expected Output:
Temperature: 40°C, Description: This command checks the temperature of
the Arista Access Node WFD-ACC-14. The output indicates whether the
temperature is within the recommended range or not.

Command: show system internal asic memory, Expected Output: ASIC
memory status: OK, Description: This command checks the ASIC memory
status of the Arista Access Node WFD-ACC-14. The output indicates
whether the ASIC memory is functioning correctly or not.
```

Roll-back Procedures

- Step: Restore the previous configuration., Command: load previous configuration, Description: This step involves restoring the previous configuration of the Arista Access Node WFD-ACC-14. The command `load previous configuration` restores the previous configuration.

- Step: Replace the original faulty component., Command: replace original faulty component, Description: This step involves replacing the original faulty component with the original one. The command `replace original faulty component` replaces the original faulty component with the original one.